Unconventional scientists chase unconventional science.

Prelude

This piece began as an exploration into a small selection of scientific discoveries that meandered uneventfully through the 20th century, slowly drying out, before returning with a surge as technologies capable of safeguarding human life against the 21st century's biological threats.

Why did it take 100 years for the true value of a scientific discovery to be realised? What were the conditions that prevented, and then liberated, these technologies from taking shape?

Whilst reading the origin stories of the humble bacteriophage and neglected glycol vapours, I came upon the scientists responsible for their entry into literature. The journey from discoveries to technologies is interesting, but the people behind them are fascinating. And so, like the science I planned to write about, this piece has evolved. Unconventional scientists chase unconventional science, and unconventional science opens up whole new fields.

Intro

Scientific progress can unfurl in a multitude of ways. An origami lotus flower - sequentially constructed fold by fold, a spluttering car engine roaring into life, or an early germination that succumbs to a bitter frost. Roots left to persist indeterminately till sunlight returns.

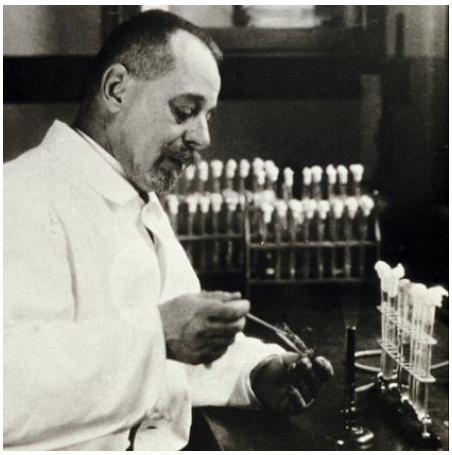
A scientist's career is made up of origami, engines, and often roots.

To carve out a career in science, I once heard the advice, '*simply follow your interests*'. But if you have many scientific interests, then what? What should one dedicate time or even a life to? In searching for a new field to couple together a background in infectious diseases with experience in implementing AI health technologies, I undertook an introductory course in Biosecurity. The recent pandemic has prompted increased focus on developing and implementing technologies to help stop the next one.

My attention was drawn to the wide-scale interventions for pandemic preparedness which leverage neglected early to mid-20th century science. The antimicrobial properties of glycol vapours had been traded out for theatre fog machines, and germicidal UV light had been presumed to be the only form of UV capable of sterilising air and surfaces, meaning the less dangerous far-UVC light was only considered from 2018. I drew instant parallels to a technology I'd previously worked on, bacteriophages. Overridden in the West by the rise of the antibiotic, the late 1910s discovery that bacteria are hunted by their very own viruses took close to a century to commercialise.

In an era in which success and impact are blurred with citation numbers, a look back at the lives of two of the pioneers of biosecurity breakthroughs uncovers how having many interests can set the work of unconventional scientists apart, even if the research begins life as roots.

The restless adventurer, discoverer of phage.



Felix d'Herelle is widely accredited as the co-discoverer of bacteriophages at the Pasteur Institute in 1917. With no university education, bar a few months of study at the University of Bonn, d'Herelle spent his late teens and early 20s exploring the world - covering Europe and South America.

Following a move to Canada from France at the age of 24, on the advice of a family friend to follow in Louis Pasteur's footprints by studying fermentation, d'Herelle became a self-taught microbiologist. His first commission was to investigate whether Canada's excess of maple syrup could be fermented into whisky.

Pasteur's career became his own guiding north star:

> *"Hunting microbes appeared to me as the last adventure in a standardised world: for a microbiologist, I took Pasteur as a model. Perhaps, I said to myself then, he had the mental training, imposed by circumstances, that played a role in the acquisition of his "microbiological mind" and therefore I should follow the same path."*

d'Herelle's adventures took him to Guatemala and Mexico to continue his study of fermentation, this time for the production of schnapps from the native sisal plant (with a detour into becoming 'the father of biopesticides' by attempting to use *Bacillus thuringiensis* to stop a locust plague). Back to France to facilitate the production of the schnapps, then onto Argentina to fight more locusts, he finally arrived at his groundbreaking phage discovery on his return to Paris at the age of 40. d'Herelle even introduced his phage research to the Soviet Union at the Tbilisi Institute, sparking the Soviet Union's enduring use of personalised phage therapies.

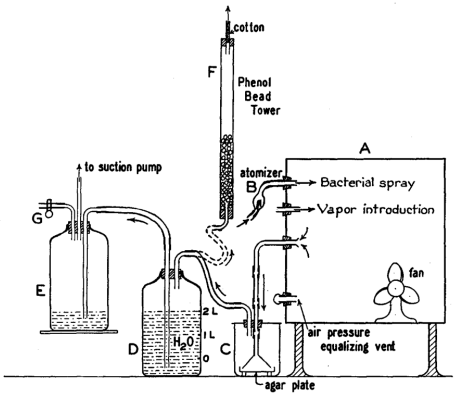
The keystone to d'Herelle's career: an ability to apply his foundational fermentation knowledge across multiple domains, all the while allowing himself to mix work and play, experimentation and mission. d'Herelle died without ever knowing the full impact of his phage research; the roots he seeded were famously nominated 10 times for the Nobel Prize, without winning once. Instead, he left behind the aptly named memoir: *Les pérégrinations d'un microbiologiste* or "Wanderings of a Microbiologist".

The catalyst scientist.

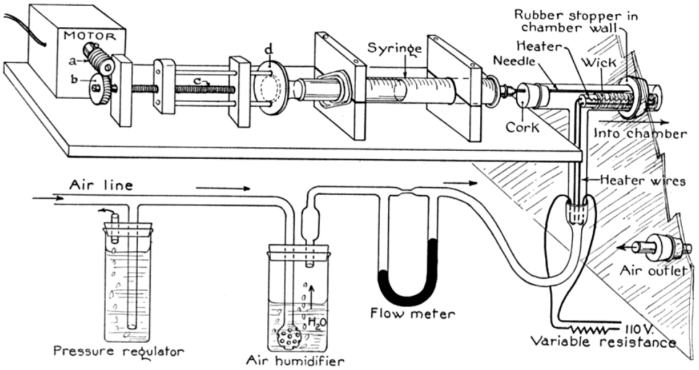
The sterilisation of air was also first investigated a century ago. Back in 1928, researchers deployed NaOCl (otherwise known as chlorine bleach) in a fine spray of electrolysed seawater and observed a significant depletion of dispersed *E. coli*.

Using bleach to eradicate airborne pathogens has a clear drawback; there is a certain irony in introducing a known dermal and respiratory irritant to control respiratory diseases. This research did, however, lay the foundations for a happy accident. The bactericidal properties of glycol vapours were discovered by a

Chicago-based team of researchers, including Theodore T. Puck, investigating bactericidal substances in aerosol form. Glycol was swapped out with water as a solvent, but was found to have a greater antimicrobial action than the substances being tested.



TEXT-FIG. 1. Apparatus employed for determining bactericidal activity of glycol vapors. *A*, 60 liter air-tight glass-walled chamber with one metal wall (opposite the side with vents) fitted with a door closing on rubber gaskets. (All sides, 15 inches square.) The three upper orifices are actually on a horizontal line across the center of the wall. *B*, Graeser atomizer connected with inlet by rubber tube. *C*, sampling jar. *G*, air vent allowing water to return from bottle *E* to *D*.



TEXT-FIG. 2. Apparatus for the introduction of air containing calculated concentrations of glycol vapor.

Puck's 1942 experimental apparatus, added in appreciation of 1940s hand-drawn figures.

The resulting paper was published in 1942; it is only in the 2020s that the antimicrobial properties of glycol vapours are being considered as a pandemic preparedness tool, thanks to Puck's groundwork. His early-career publications are indicative of an unsung pioneer and champion of biosecurity research long before the term was coined. Following five years diligently studying the mechanisms of glycol vapour air disinfection, Puck pivoted entirely and spent a further five years working on none other than phage(!).

A move to the University of Colorado, Denver, marked a career-making achievement for Puck as the first researcher to clone HeLa cells - a completely new domain for him. He also developed the CHO cell line from Chinese hamster ovarian cells. Then, in 1960, his lab produced unequivocal evidence that the human genome consists of 46 chromosomes, contrary to the 48 assumed at the time.

Across his career, Puck made significant contributions to not one but three fields, pursuing origami, engines, and roots. He remained modest about his achievements and did not live to see the potential impact of his glycol vapour discovery. In contrast to d'Herelle, he was a man dedicated to his lab at Colorado - his obituary describes a scientist who, at the age of 89, continued to perform experiments till a few weeks before his death. Following his interests.

To conclude.

Did d'Herelle and Puck have suspicion of the gravity of what they were working on across their lifetimes? Or were they just following their interests? Neither man's career was marked by external validation, but both inexplicably changed the course of science. Their careers were fluid and interests wide, the thread of commonality: commit to what is in front of you at this very moment in time. In the search for a field, maybe it is best to just follow your interests.